

3. Klausur PraMa

Gruppe B

Aufgabe 1

$$y \sin(xy) - 2 + (x \cdot \sin(xy) + 10) \frac{dy}{dx} = 0$$

$$(y \cdot \sin(xy) - 2) dx + (x \cdot \sin(xy) + 10) dy = 0$$

$$a) \left. \begin{aligned} \frac{d}{dy} (y \cdot \sin(xy) - 2) &= \sin(xy) + y \cdot \cos(xy) \cdot x \\ \frac{d}{dx} (x \cdot \sin(xy) + 10) &= \sin(xy) + x \cdot \cos(xy) \cdot y \end{aligned} \right\} =$$

IB erfüllt $\Rightarrow$ exakt

$$b) \Phi = \int (y \cdot \sin(xy) - 2) dx = -\frac{1}{y} \cdot y \cdot \cos(xy) - 2x + c(y)$$

$$= -\cos(xy) - 2x + c(y)$$

NR:

$$\int \sin(xy) dx = \left| \begin{array}{l} u = xy \\ du = y dx \\ \frac{1}{y} du = dx \end{array} \right| = \frac{1}{y} \cdot \int \sin(u) du = -\frac{1}{y} \cos(xy)$$

$$\frac{d\Phi}{dy} = \sin(xy) \cdot x + c'(y)$$

$$x \cdot \sin(xy) + c'(y) \stackrel{!}{=} x \cdot \sin(xy) + 10$$

$$c'(y) = 10$$

$$c(y) = \int 10 dy = 10y + c_0$$

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12307040

$$\Phi = -\cos(xy) - 2x + 10y + c_0$$

Aufgabe 2

$$x \cdot u_x = (1+t) u_t$$

$$u(x, 0) = 4x^3 + x^2$$

Separationsansatz:

$$u(x, t) = v(t) w(x)$$

$$x \cdot v(t) w'(x) = (1+t) v(t) w(x) \quad | : (v(t) w(x))$$

$$x \cdot \frac{w'(x)}{w(x)} = (1+t) \cdot \frac{v(t)}{v(t)} = \mathcal{K}$$

$$1) \quad x \cdot \frac{w'}{w} = \mathcal{K}$$

$$\frac{w'}{w} = \mathcal{K} \cdot \frac{1}{x}$$

$$\frac{1}{w} \frac{dw}{dx} = \mathcal{K} \cdot \frac{1}{x}$$

$$\int \frac{1}{w} dw = \mathcal{K} \cdot \int \frac{1}{x} dx$$

$$\ln|w| = \mathcal{K} \cdot \ln|x| + c_1 \quad | \exp()$$

$$w(x) = c_1 \cdot x^\kappa$$

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12307040

$$2) (1+t) \cdot \frac{\dot{v}(t)}{v(t)} = \kappa$$

$$\frac{\dot{v}}{v} = \kappa \cdot \frac{1}{1+t}$$

$$\frac{1}{v} \frac{dv}{dt} = \kappa \cdot \frac{1}{1+t}$$

$$\int \frac{1}{v} dv = \kappa \cdot \int \frac{1}{1+t} dt$$

$$\ln|v| = \kappa \cdot \ln|1+t| + c_2$$

$$\ln|v| = \ln|(1+t)^\kappa| + c_2 \quad / \exp()$$

$$v(t) = c_2 \cdot (1+t)^\kappa$$

$$u(x,t) = c_1 \cdot x^\kappa \cdot c_2 \cdot (1+t)^\kappa$$

$$u(x,t) = c \cdot x^\kappa \cdot (1+t)^\kappa$$

$$u(x,0) = 4x^3 + x^2$$

$$u(x,0) = c \cdot x^\kappa \cdot 1^\kappa = c \cdot x^\kappa$$

Superposition:

$$i) c=4, \kappa=3 \quad ii) c=1, \kappa=2$$

$$u(x,t) = 4x^3 \cdot (1+t)^3 + x^2 \cdot (1+t)^2$$

Aufgabe 3

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12307040

$$t^3 \ddot{y} + 3t^2 \dot{y} - 3ty = 3$$

$$a) \quad t^3 \ddot{y} + 3t^2 \dot{y} - 3ty = 0 \quad | :t$$
$$t^2 \ddot{y} + 3t \dot{y} - 3y = 0$$

$$y = t^\alpha, \quad \dot{y} = \alpha \cdot t^{\alpha-1}, \quad \ddot{y} = \alpha \cdot (\alpha-1) \cdot t^{\alpha-2}$$
$$= (\alpha^2 - \alpha) \cdot t^{\alpha-2}$$

$$t^2 \cdot t^{\alpha-2} \cdot (\alpha^2 - \alpha) + 3t \cdot t^{\alpha-1} \cdot \alpha - 3 \cdot t^\alpha = 0$$

$$t^\alpha \cdot (\alpha^2 - \alpha) + 3\alpha t^\alpha - 3t^\alpha = 0$$

$$t^\alpha \cdot (\alpha^2 - \alpha + 3\alpha - 3) = 0$$

$$t^\alpha \cdot (\alpha^2 + 2\alpha - 3) = 0$$

$$\underbrace{\hspace{10em}}_{=0}$$

$$\alpha^2 + 2\alpha - 3 = 0$$

$$\alpha_{1,2} = -\frac{2}{2} \pm \sqrt{1^2 + 3}$$

$$\alpha_{1,2} = -1 \pm \sqrt{4}$$

$$\alpha_{1,2} = -1 \pm 2$$

$$\alpha_1 = -3, \quad \alpha_2 = 1$$

$$y_h(t) = c_1 \cdot t^{-3} + c_2 \cdot t$$

$$b) \quad y = c_1 t^{-3} + c_2 t$$

$$\dot{y} = \dot{c}_1 t^{-3} - 3c_1 t^{-4} + \dot{c}_2 t + c_2$$

Wir fordern:

$$\dot{c}_1 t^{-3} + \dot{c}_2 t = 0 \Rightarrow \dot{c}_2 t = -\dot{c}_1 t^{-3}$$

$$\dot{c}_2 = -\dot{c}_1 t^{-4}$$

$$\dot{y} = -3c_1 t^{-4} + c_2$$

$$\ddot{y} = -3\dot{c}_1 t^{-4} + 12c_1 t^{-5} + \dot{c}_2$$

$$t^2 \ddot{y} + 3t \dot{y} - 3y = 3$$

$$i) \quad t^2 \cdot (-3\dot{c}_1 t^{-4} + 12c_1 t^{-5} + \dot{c}_2) =$$

$$= -3\dot{c}_1 t^{-2} + \cancel{12c_1 t^{-3}} + \dot{c}_2 t^2$$

$$ii) \quad 3t \cdot (-3c_1 t^{-4} + c_2) =$$

$$= \cancel{-9c_1 t^{-3}} + \cancel{3c_2 t}$$

$$iii) \quad -3 \cdot (c_1 t^{-3} + c_2 t) =$$

$$= \cancel{-3c_1 t^{-3}} - \cancel{3c_2 t}$$

$$-3\dot{c}_1 t^{-2} + \dot{c}_2 t^2 = 3$$

aus Forderung:

$$\dot{c}_2 = -\dot{c}_1 t^{-4}$$

Daniel Köstler

12307040

$$-3\dot{c}_1 t^{-2} - \dot{c}_1 t^{-4} t^2 = 3$$

$$-3\dot{c}_1 t^{-2} - \dot{c}_1 t^{-2} = 3$$

$$-4\dot{c}_1 t^{-2} = 3$$

$$-4\dot{c}_1 = 3t^2$$

$$\dot{c}_1 = -\frac{3}{4} t^2$$

$$c_1 = -\frac{3}{4} \int t^2 dt$$

$$c_1 = -\frac{3}{4} \frac{t^3}{3} = -\frac{3}{12} t^3 = -\frac{1}{4} t^3$$

$$\dot{c}_2 = -\dot{c}_1 t^{-4}$$

$$\dot{c}_2 = \frac{3}{4} t^2 t^{-4} = \frac{3}{4} t^{-2}$$

$$c_2 = \frac{3}{4} \int t^{-2} dt = \frac{3}{4} \frac{t^{-1}}{-1} = -\frac{3}{4} t^{-1}$$

$$y = c_1 t^{-3} + c_2 t$$

$$y_p = -\frac{1}{4} t^3 t^{-3} - \frac{3}{4} t^{-1} t^1 = -\frac{1}{4} - \frac{3}{4} = -1$$

$$y(t) = c_1 t^{-3} + c_2 t - 1$$

$$\begin{aligned} c) \quad t^3 \ddot{y} + 3t^2 \dot{y} - 3ty &= 3 \quad | : t^3 \\ \ddot{y} + 3t^{-1} \dot{y} - 3t^{-2} y &= 3t^{-3} \\ \ddot{y} &= -3t^{-1} \dot{y} + 3t^{-2} y + 3t^{-3} \end{aligned}$$

$$\begin{aligned} x_1 &= y & \dot{x}_1 &= \dot{y} = x_2 \\ x_2 &= \dot{y} & \dot{x}_2 &= \ddot{y} \end{aligned}$$

$$\ddot{y} = -3t^{-1} x_2 + 3t^{-2} x_1 + 3t^{-3}$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -3t^{-1} x_2 + 3t^{-2} x_1 + 3t^{-3} \end{pmatrix}$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 3t^{-2} & -3t^{-1} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 3t^{-3} \end{pmatrix}$$

$$\dot{\vec{x}}(t) = \begin{pmatrix} 0 & 1 \\ 3t^{-2} & -3t^{-1} \end{pmatrix} \cdot \vec{x}(t) + \begin{pmatrix} 0 \\ 3t^{-3} \end{pmatrix}$$

$$y(t) = y_h(t) + y_p(t)$$

$$y_h(t) = c_1 y_1(t) + c_2 y_2(t)$$

$$Y(t) = \begin{pmatrix} y_1 & y_2 \\ \dot{y}_1 & \dot{y}_2 \end{pmatrix} \quad \det Y \neq 0 \dots \text{p.u.}$$

$\{y_1, y_2\} \dots$ Fundamentalsystem

Wellengleichung

$$c^2 \cdot u_{xx} = u_{tt}$$

$$u(t, x) = f(x-ct), \quad u(t, x) = f(x+ct)$$

$$u(t, x) = f(x-ct) + g(x+ct)$$

Separationsansatz

$$u(x, t) = v(t)w(x); \quad \frac{\dot{v}(t)}{v(t)} = \lambda = \frac{w''(x)}{w(x)} \\ \lambda = -\alpha^2$$

Euler DGL

$$a_2 t^2 y'' + a_1 t y' + a_0 y = 0$$

$$\text{Ansatz: } y(t) = t^\alpha$$

Exakte DGL

$$p(x, y) dx + q(x, y) dy = 0$$

$$\frac{dp}{dy} = \frac{dq}{dx} \dots \text{exakt}$$

Bernoulli DGL

$$\dot{y} = a(t)y + b(t)y^\alpha$$

$$\text{Ansatz: } y = z^{\frac{1}{1-\alpha}}$$

Systeme von DGL

$$x_1 = y, \quad x_2 = \dot{y}, \dots$$

$$\dot{x}_1 = \dot{y} = x_2, \quad \dot{x}_2 = \ddot{y}, \dots$$

$$\dot{\vec{x}} = A \cdot \vec{x} + \vec{b}$$

$$A = T D T^{-1}$$

$$\underbrace{T^{-1} \dot{\vec{x}}}_{\dot{\vec{z}}} = \underbrace{D T^{-1} \vec{x}}_{\vec{z}}$$

$$\dot{\vec{z}} = T^{-1} \dot{\vec{x}} \Rightarrow \vec{x} = T \vec{z}$$

Ricatti DGL

$$\dot{y} = a(t)y + b(t)y^2 + r(t)$$

$$\dot{z} = b z^2 + (2y_p b + a) \cdot z$$

$$y(t) = y_p + z$$

$\Rightarrow$ Bernoulli

f	f'	f	F
$\sin x$	$\cos x$	0	C
$\cos x$	$-\sin x$	$\frac{1}{x}$	$\ln x $
$\tan x$	$\frac{1}{\cos^2 x}$	$\sin x$	$-\cos x$
$\cot x$	$-\frac{1}{\sin^2 x}$	$\cos x$	$\sin x$
$\ln x$	$\frac{1}{x}$	$\tan x$	$-\ln \cos x $
$\cosh x$	$\sinh x$	a^x	$\frac{a^x}{\ln a}$
$\sinh x$	$\cosh x$	$\ln x$	$x \cdot \ln x - x$
		$\log_a(x)$	$\frac{1}{\ln a} \cdot (x \cdot \ln x - x)$

Ableitungsregeln

$$(f_1 \cdot f_2)' = f_1' \cdot f_2 + f_1 \cdot f_2'$$

$$\left(\frac{f_1}{f_2}\right)' = \frac{f_1' \cdot f_2 - f_1 \cdot f_2'}{f_2^2}$$

Integrationsregeln

$$\int a(x)b(x)dx = a(x) \cdot d - \int c \cdot d dx$$

$$(a(x))' = c$$

$$\int b(x)dx = d$$

Trigonometrie

$$\sin^2(x) + \cos^2(x) = 1 \quad \sin(2x) = 2 \cdot \sin(x) \cdot \cos(x)$$

$$\cos(2x) = 1 - 2\sin^2(x) \quad \cos(2x) = 2\cos^2(x) - 1 \quad \cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\sin(x) \cdot \sin(y) = \frac{1}{2} \cdot (\cos(x-y) - \cos(x+y))$$

$$\cos(x) \cdot \cos(y) = \frac{1}{2} \cdot (\cos(x-y) + \cos(x+y))$$

$$\sin(x) \cdot \cos(y) = \frac{1}{2} \cdot (\sin(x-y) + \sin(x+y))$$

$$\cosh^2(x) - \sinh^2(x) = 1$$

$$\sinh(x) = \frac{1}{2} \cdot (e^x - e^{-x})$$

$$\cosh(x) = \frac{1}{2} \cdot (e^x + e^{-x})$$

Linearisierung

$$T_1(\varphi) = f(\varphi_0) + f'(\varphi_0) \cdot (\varphi - \varphi_0)$$

$$T_0(\varphi) = f(\varphi_0)$$

Große Lösungsformel

$$ax^2 + bx + c = 0$$

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Inhomogenität f	Ansatz für y_p
$f(t) = b$	$y_p = a$
$f(t) = be^{\lambda t}$	$y_p = ae^{\lambda t}$
$f(t) = b_0 + b_1 t + \dots + b_m t^m$	$y_p = a_0 + a_1 t + \dots + a_m t^m$
$f(t) = b_1 \sin(kt) + b_2 \cos(kt)$	$y_p = a_1 \sin(kt) + a_2 \cos(kt)$
$f(t) = e^{\lambda t} (b_1 \sin(kt) + b_2 \cos(kt))$	$y_p = e^{\lambda t} (a_1 \sin(kt) + a_2 \cos(kt))$
$f(t) = e^{\lambda t} \cdot p(t)$	$y_p(t) = e^{\lambda t} q(t)$

$$e^{i\varphi} = \cos(\varphi) + i \cdot \sin(\varphi)$$

$$\ln(1) = 0$$

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}$$